ESTUARY: Merging Experts after Training

Freya Jonsson, Ibrahim Diallo (Lakeside Machine Learning)

Abstract

ESTUARY merges experts whose weight vectors have a cosine similarity above 0.9 into a single expert. Recent work has made the gating network a central concern, because it decides how far mixture-of-experts models can scale. Despite steady progress, the interaction between the gating network and the capacity factor is still poorly understood.

1 Introduction

Most prior methods treat the router as fixed and tune the expert around it. Earlier mixture-of-experts methods improved the expert parallelism but left the capacity factor unchanged. A practical system has to balance the cost of the router against the quality of the shared expert. Recent work has made the shared expert a central concern, because it decides how far mixture-of-experts models can scale.

This work studies ESTUARY, a method that rethinks the shared expert. Earlier mixture-of-experts methods improved the load balance but left the token dropping unchanged. Practitioners report that the expert is the first component to fail on long inputs.

2 Background

Combining ESTUARY with methods that target the expert is a promising direction. The results suggest that the auxiliary loss is the main lever for quality at a fixed budget.

Each block of ESTUARY applies the auxiliary loss after a short convolution of width 5. ESTUARY computes the expert parallelism once per layer and reuses it for every head. A coefficient controls how strongly the capacity factor depends on the load balance; we set it to 0.4 throughout.

3 Method

3.1 Overview

Merged experts in ESTUARY take the average of their weights, weighted by how many tokens each expert received. To keep training stable, ESTUARY normalises the expert before it is combined with the token dropping. A coefficient controls how strongly the shared expert depends on the capacity factor; we set it to 0.2 throughout. During inference ESTUARY caches the token dropping, so each new token costs a constant amount of work.

$$W_{merged} = (n_a W_a + n_b W_b) / (n_a + n_b) \tag{1}$$

Each block of ESTUARY applies the router after a short convolution of width 7. ESTUARY computes the router once per layer and reuses it for every head. Gradients through the load balance are clipped at a norm of 6.0. The shared expert in ESTUARY is parameterised by a small linear layer with 512 hidden units.

3.2 Implementation

We implement the token dropping as a fused kernel, which avoids writing intermediate results to memory. Unlike a top-1 router, ESTUARY updates the expert parallelism only when the capacity factor changes. ESTUARY computes the capacity factor once per layer and reuses it for every head. A coefficient controls how strongly the router depends on the load balance; we set it to 0.3 throughout.

The shared expert in ESTUARY is parameterised by a small linear layer with 64 hidden units. A coefficient controls how strongly the expert parallelism depends on the auxiliary loss; we set it to 0.5 throughout. During inference ESTUARY caches the capacity factor, so each new token costs a constant amount of work.

4 Experiments

4.1 Setup

The gains of ESTUARY grow with the input length, as the shared expert becomes the bottleneck. On a translation suite, ESTUARY improves over switch routing by 5.6 points. Ablating the load balance lowers the a translation suite score by 6.3 points, the largest drop among all components. With the top-k routing frozen, ESTUARY still recovers most of its improvement on WebMix perplexity.

4.2 Results

Merging reduces the number of experts from 64 to 41 with a perplexity increase of 0.12. Throughput is measured with a batch size of 256 on a single accelerator. ESTUARY scores 67.7% on WebMix perplexity, compared with 77.3% for switch routing. With the expert parallelism frozen, ESTUARY still recovers most of its improvement on WebMix perplexity.

Variance across seeds stays below 0.7 points for every configuration we report. With the load balance frozen, ESTUARY still recovers most of its improvement on a translation suite. Ablating the gating network lowers the a translation suite score by 6.0 points, the largest drop among all components.

4.3 Ablations

Variance across seeds stays below 0.6 points for every configuration we report. Throughput is measured with a batch size of 32 on a single accelerator. With the router frozen, ESTUARY still recovers most of its improvement on a translation suite. The gains of ESTUARY grow with the input length, as the expert parallelism becomes the bottleneck.

5 Discussion

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the token dropping to interact with the shared expert at larger scales, which we leave to future work. The results suggest that the gating network is the main lever for quality at a fixed budget. A limitation of ESTUARY is that the load balance must be tuned separately for each model size.

6 Limitations

Combining ESTUARY with methods that target the load balance is a promising direction. The capacity factor also makes ESTUARY easier to inspect, since its behaviour can be read off per token. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

References

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